Supplementary material for: Drift-aware meta-adaptive anomaly detection for few-shot, non-stationary industrial IoT streams: a benchmarking protocol and the DAMAD detector

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S1. The review corpus

Table S1. The 66 papers of the systematic review in Section 3 of the main paper, with the codes of the four taxonomy axes (D1 paradigm, D2 mechanism, D3 streaming capability, D4 validation; abbreviations as in Section 3.2). The screening sheet with every excluded record and its reason is released with the code.

#	Year	First author	Title	Venue	D1	D2	D3	D4	DOI
1	2020	Shen Zhang	Few-Shot Bearing Anomaly Detection Based on Model-Agnostic Meta-Learning	2020 International Conference on Electrical Machines and Sys	OPT	CLS	STA	IOV	10.23919/ICEMS50442.2020.9291099
2	2021	Jinglin Zhang	Diagnosis of Interturn Short-Circuit Faults in Permanent Magnet Synchronous Motors Based on Few-Shot Learning Under a Federated Learning Framework	IEEE Transactions on Industrial Informatics	MET	DIS	STA	IOV	10.1109/tii.2021.3067915
3	2021	Xiaokang Zhou	Siamese Neural Network Based Few-Shot Learning for Anomaly Detection in Industrial Cyber-Physical Systems	IEEE Transactions on Industrial Informatics	MET	DIS	STA	IOV	10.1109/tii.2020.3047675
4	2021	G. Yu	Subspace-based domain adaptation for few-shot fault diagnosis	2021 CAA Symposium on Fault Detection, Supervision, and Safe	NML	CLS	STA	IOV	10.1109/safeprocess52771.2021.96936
5	2022	Mohamed-Ali Tnani	Efficient Feature Learning Approach for Raw Industrial Vibration Data Using Two-Stage Learning Framework.	Sensors (Basel, Switzerland)	MET	DIS	STA	IOV	10.3390/s22134813
6	2022	Yi-Cheng Yu	Fast Adaption for Multi Motor Anomaly Detection via Meta Learning	2022 IEEE 31st International Symposium on Industrial Electro	OPT	REC	STA	IOV	10.1109/isie51582.2022.9831559

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			and deep unsupervised learning						
7	2022	Y. Liu	The Adaptive Personalized Federated Meta-Learning for Anomaly Detection of Industrial Equipment	IEEE Journal of Radio Frequency Identification	OPT	CLS	STA	IOV	10.1109/jrfid.2022.3209657
8	2022	Shreshth Tuli	TranAD	Proceedings of the VLDB Endowment	OPT	REC	STA	IIV	10.14778/3514061.3514067
9	2022	T. Ye	FLAG: Few-Shot Latent Dirichlet Generative Learning for Semantic-Aware Traffic Detection	IEEE Transactions on Network and Service Management	NML	CLS	STA	IOV	10.1109/tnsm.2021.3131266
10	2023	Moon Noh Jung Lee Hwang	Anomaly detection using a model-agnostic meta-learning-based variational auto-encoder for facility management	Journal of Building Engineering	OPT	REC	STA	GEN	10.1016/j.jobe.2023.106099
11	2023	Farhan Md. Siraj	A Few-Shot Learning Based Fault Diagnosis Model Using Sensors Data from Industrial Machineries	Vibration	MET	DIS	STA	IOV	10.3390/vibration6040059
12	2023	Chandana Priya Nivarthi	Towards Few-Shot Time Series Anomaly Detection with Temporal Attention and Dynamic Thresholding	2023 International Conference on Machine Learning and Applic	NML	REC	STA	GEN	10.1109/icmla58977.2023.00218
13	2023	Ke Yue	An Emerging Fault Diagnosis Framework with Incremental Meta-Learning for Rotating Machinery	2023 International Conference on Sensing, Measurement & Data	OPT	CLS	INC	IOV	10.1109/icsmd60522.2023.10490806
14	2023	L. Cheng	MMFSL: A Novel Multimodal Few-Shot Learning Framework for Fault Diagnosis of Industrial Bearings	IEEE Transactions on Instrumentation and Measurement	NML	CLS	STA	IOV	10.1109/tim.2023.3289549

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15	2023	Caiyun Wu	Engine fault detection algorithm EA-EFNC based on few-shot learning	Third International Conference on Machine Learning and Compu	NS	CLS	STA	IOV	10.1117/12.2675146
16	2023	V. L. Cao	Few-Shot Learning with Discriminative Representation for Cyberattack Detection	2023 15th International Conference on Knowledge and Systems	MET	DIS	STA	IOV	10.1109/kse59128.2023.10299444
17	2023	Z. Shi	Few-Shot Network Intrusion Detection Based on Model-Agnostic Meta-Learning with L2F Method	2023 IEEE Wireless Communications and Networking Conference	OPT	CLS	STA	IOV	10.1109/wcnc55385.2023.10118898
18	2023	X. Ye	Rolling bearing fault diagnosis with variable load and few samples based on multifeature fusion meta-learning	2023 CAA Symposium on Fault Detection, Supervision and Safet	OPT	CLS	STA	IOV	10.1109/safeprocess58597.2023.10295
19	2023	J. Chen	A Meta-Learning Method for Electric Machine Bearing Fault Diagnosis Under Varying Working Conditions With Limited Data	IEEE Transactions on Industrial Informatics	OPT	CLS	STA	IOV	10.1109/tii.2022.3165027
20	2023	Y. Liu	Semi-supervised Few-shot Network Intrusion Detection based on Meta-learning	2023 IEEE International Conferences on Internet of Things (i	NS	CLS	STA	IOV	10.1109/ithings-greencom-cpscom-smartdata-cybermatics60724.2023.000
21	2023	S. Fang	Few-Shot Fault Diagnosis of Rolling Bearings Based on Multidimensional Global Attention Siamese Network	IECON 2023-49th Annual Conference of the IEEE Industrial El	MET	DIS	STA	IOV	10.1109/iecon51785.2023.10312375
22	2023	X. Zhang	Imbalanced Fault Diagnosis Based on Supervised Prototypical Contrastive Network	2023 Global Reliability and Prognostics and Health Managemen	MET	DIS	STA	IOV	10.1109/phm-hangzhou58797.2023.10482577
23	2024	Yan et al.	Meta learning-based few-shot intrusion detection for 5G-	Complex & Intelligent Systems	MET	DIS	STA	IOV	10.1007/s40747-024-01388-1

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			enabled industrial internet						
24	2024	Lang Du	Research on Turnout Fault Diagnosis Model Based on GADF-CNN-MAML	2024 5th International Conference on Artificial Intelligence	OPT	CLS	STA	IOV	10.1109/icaice63571.2024.10863959
25	2024	Wang Danfeng	Model-Agnostic Meta-Learning-based Fault Classification for Industrial Processes with Small Sample	2024 39th Youth Academic Annual Conference of Chinese Associ	OPT	CLS	STA	IOV	10.1109/yac63405.2024.10598657
26	2024	P. Jagdish Kumar	Classifying Network Abnormalities Into Faults and Attacks in Iot-Based Cyber-Physical Systems using Machine Learning	2024 International Conference on Integrated Circuits and Com	MET	DIS	STA	IOV	10.1109/icicacs60521.2024.10498664
27	2024	Yi-Cheng Yu	Pruning Quantized Unsupervised Meta-Learning DegradingNet Solution for Industrial Equipment and Semiconductor Process Anomaly Detection and Prediction	Applied Sciences	NS	REC	STA	IOV	10.3390/app14051708
28	2024	Yunpu Wu	A Novel Fault Diagnosis Method of High-Speed Train Based on Few-Shot Learning.	Entropy (Basel, Switzerland)	NS	CLS	STA	IOV	10.3390/e26050428
29	2024	J. Wu	A Novel Few-Shot Deep Transfer Learning Method for Anomaly Detection: Deep Domain-Adversarial Contrastive Network With Time-Frequency Transferability Analytics	IEEE Internet of Things Journal	NML	CLS	STA	IOV	10.1109/jiot.2024.3403976
30	2024	L. A. Jilcha	SaRLog: Semantic-Aware Robust Log Anomaly Detection via BERT-	IEEE Internet of Things Journal	MET	DIS	STA	GEN	10.1109/jiot.2024.3386183

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			Augmented Contrastive Learning						
31	2024	Z. Pu	Incrementally Generative Adversarial Diagnostics Using Few-Shot Enabled One-Class Learning	IEEE Transactions on Industrial Informatics	NML	HYB	INC	IOV	10.1109/tii.2024.3416945
32	2024	H. Sun	Space Decoupled Prototype Learning for Few-Shot Attack Detection in Cyber–Physical Systems	IEEE Transactions on Industrial Informatics	MET	DIS	STA	IOV	10.1109/tii.2024.3423327
33	2024	U. Zukaib	Meta-IDS: Meta-Learning-Based Smart Intrusion Detection System for Internet of Medical Things (IoMT) Network	IEEE Internet of Things Journal	NS	CLS	STA	IOV	10.1109/jiot.2024.3387294
34	2024	K. Li	Reweighted Regularized Prototypical Network for Few-Shot Fault Diagnosis	IEEE Transactions on Neural Networks and Learning Systems	MET	DIS	STA	IOV	10.1109/tnnls.2022.3232394
35	2024	L. Du	A Few-Shot Class-Incremental Learning Method for Network Intrusion Detection	IEEE Transactions on Network and Service Management	NS	CLS	INC	IOV	10.1109/tnsm.2023.3332284
36	2025	Jiaxi Li	An Anomaly Detection Approach for Aerospace Telemetry Data Based on Model-Agnostic Meta-Learning	IFAC-PapersOnLine	OPT	REC	STA	IOV	10.1016/j.ifacol.2025.11.424
37	2025	Yutian Wang	Few-Shot Learning for Anomaly Detection in Gas-Fired Power Plants Using Prototypical Networks	Informatica	MET	DIS	STA	IOV	10.31449/inf.v49i36.12350
38	2025	N. Ijaz	A Hybrid LATAM and Few-Shot Learning Framework for Fault Diagnosis	IEEE Sensors Journal	MET	HYB	STA	IOV	10.1109/jsen.2025.3620491

#	Year	First author	Title	Venue	D1	D2	D3	D4	DOI
			in Wireless Sensor Networks						
39	2025	Ming Wan	One-Class Anomaly Detection for Industrial Data Traffic Based on Dual-Characteristic Prototype Points	2025 IEEE International Symposium on Parallel and Distribute	MET	DIS	STA	IOV	10.1109/ispa67752.2025.00089
40	2025	David He	Full Ceramic Bearing Fault Diagnosis with Few-Shot Learning Using GPT-2	Computer Modeling in Engineering & Sciences	MOD	CLS	STA	IOV	10.32604/cmes.2025.063975
41	2025	Ravindra Chopparapu	Implementation of an Optimized Hybrid Machine Learning–Based IDS for Enhanced Threat Detection in IoT Environments	2025 IEEE 5th International Conference on ICT in Business In	NS	CLS	STA	IOV	10.1109/ictbig68706.2025.11323664
42	2025	Ali Pourghoraba	Model-Agnostic Meta-Learning for Fault Diagnosis of Induction Motors in Data-Scarce Environments with Varying Operating Conditions and Electric Drive Noise	IEEE Transactions on Energy Conversion	OPT	CLS	STA	IOV	10.1109/tec.2025.3556100
43	2025	Ying Yu	A novel few-shot cross-domain fault diagnosis method enhanced by feature fusion and ReLU-KAN	Engineering Research Express	NS	CLS	STA	IOV	10.1088/2631-8695/adcc77
44	2025	Minke Zou	HySAM: A Hybrid Similarity-Enhanced Attention-Driven Network for Multi-Scale Few-Shot Learning in Belt Conveyor Roller Fault Detection	Proceedings of the 4th International Conference on Computer,	MET	DIS	STA	IOV	10.1145/3727648.3727708
45	2025	Wei Gao	Metric-Based Meta-Learning Relation Network for Cross-Domain Few-Shot	IEEE Sensors Journal	MET	DIS	STA	IOV	10.1109/jsen.2025.3547258

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			Bearing Fault Diagnosis						
46	2025	Yuanfa Cen	Integration of Normalizing Flows and Prototypical Networks for Few-Shot Early Fault Detection in Transformer	2025 5th International Conference on Electrical Engineering	HYB	HYB	STA	IOV	10.1109/ic2ecs68700.2025.11362286
47	2025	Yanfei Liu	A novel temporal classification prototype network for few-shot bearing fault detection.	Scientific reports	MET	DIS	STA	IOV	10.1038/s41598-025-98963-4
48	2025	Ji-Min Woo	Meta-Learning-Based LSTM-Autoencoder for Low-Data Anomaly Detection in Retrofitted CNC Machine Using Multi-Machine Datasets	Systems	OPT	REC	STA	IOV	10.3390/systems13070534
49	2025	Muyan Yao	Rethinking Discrepancy Analysis: Anomaly Detection via Meta-Learning Powered Dual-Source Representation Differentiation	IEEE Transactions on Automation Science and Engineering	NS	REC	STA	IOV	10.1109/tase.2024.3486688
50	2025	S. Zhu	Partially-Supervised Graph Derivation Network With Meta-Learning for Time-Series Anomaly Detection	IEEE Internet of Things Journal	OPT	HYB	STA	GEN	10.1109/jiot.2025.3558273
51	2025	O. S. Ogunleye	Hybrid Meta-Deep Learning Approach for Malicious Data Classification in Iot Packet Data Networks	2025 8th Conference on Cloud and Internet of Things (CIoT)	MET	DIS	STA	IOV	10.1109/ciot67574.2025.11410126
52	2025	Y. Wang	Fault Diagnosis for Pneumatic Control Valves Using a Novel Few-shot Learning Method	2025 CAA Symposium on Fault Detection, Supervision, and Safe	OPT	CLS	STA	IOV	10.1109/safeprocess67117.2025.11267
53	2025	N. Ijaz	Few-Shot Transfer Learning-Based	IEEE Access	NML	CLS	STA	IOV	10.1109/access.2025.3554748

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			Fault Classification in Wireless Sensor Networks						
54	2025	M. Zabin	Few-Shot Learning-Based Machine Fault Diagnosis Using EMD-Gammatone Spectrogram with Limited Labeled Audio Dataset	2025 IEEE International Conference on Big Data and Smart Com	NML	CLS	STA	IOV	10.1109/bigcomp64353.2025.00046
55	2025	B. Chandrasekaran	Evaluation of Few-Shot Learning with Vision Language Models for Needle Bearing Defect Detection	SoutheastCon 2025	MOD	CLS	STA	IOV	10.1109/southeastcon56624.2025.1097
56	2025	K. Yi	Pipeline Anomaly Detection Based on KPCA and Cosine Distance Prototypical Network	2025 IEEE International Conference on Industrial Technology	MET	DIS	STA	IOV	10.1109/icit63637.2025.10965214
57	2025	U. Suleymanov	SSPNet: Semi-Supervised Prototypical Networks for Few-Shot Network Intrusion Detection	MILCOM 2025 - 2025 IEEE Military Communications Conference (	MET	DIS	STA	IOV	10.1109/milcom64451.2025.11310333
58	2025	M. Emadaleslami	Differential Fault Diagnosis in Power Transformers Under Limited Data Condition via Few-Shot Learning	IEEE Transactions on Industrial Informatics	NS	CLS	STA	IOV	10.1109/tii.2025.3586057
59	2025	H. Li	A Few-Shot Network Attack Detection Method Based on MAML-L2F Dynamic Parameter Optimization	2025 China Automation Congress (CAC)	OPT	CLS	STA	IOV	10.1109/cac67268.2025.11486720
60	2025	J. Tian	OMLog: Online Log Anomaly Detection for Evolving System With Meta-Learning	IEEE Internet of Things Journal	MOD	PRE	DAW	GEN	10.1109/jiot.2025.3569628
61	2025	R. Xu	Cyber Attack Detection for	2025 10th International	NML	CLS	STA	IIV	10.1109/crc67523.2025.11288189

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			Industrial Control Systems with Few-Shot and Class Imbalance: An Improved DA-LSTM-UNet Scheme	Conference on Control, Robotics and					
62	2025	W. Xi	MHT: Dual-Layered Meta-Driven Hypernetworks for Unknown Attack Detection	2025 10th International Conference on Computer and Communica	MOD	CLS	STA	IOV	10.1109/icccs65393.2025.11069655
63	2025	D. N. Chi Nam	STRIDE: Spectral-Temporal Representation and Instance-Weighted Domain Expansion	2025 IEEE International Conference on Data Mining (ICDM)	NML	CLS	STA	IOV	10.1109/icdm65498.2025.00125
64	2025	J. Li	MTSNN: A Few-Shot Fine-Grained Diagnosis Framework for Cross-Machine Fault Diagnosis With Heterogeneous Fault Categories	IEEE Transactions on Instrumentation and Measurement	MET	DIS	STA	IOV	10.1109/tim.2025.3551907
65	2025	J. Xu	A Novel Approach for Small-Sample Open-Set Fault Diagnosis in Rolling Bearings: Dense-ProtoNet-AE	2025 CAA Symposium on Fault Detection, Supervision, and Safe	HYB	HYB	STA	IOV	10.1109/safeprocess67117.2025.11268
66	2025	W. Guo	ModelAdaptive Bearing Fault Diagnosis in Small Sample Situations	2025 CAA Symposium on Fault Detection, Supervision, and Safe	OPT	CLS	STA	IOV	10.1109/safeprocess67117.2025.11267